

MAE 140 Winter 12 Homework #4

T R & T, 3.9, 3.10, 3.12, 3.15 a) and c), 3.20 a) and b), 3.22, 3.26, 4.2, 4.6, 4.13

Due: 02/14/2012

3-9 (a) Formulate mesh-current equations for the circuit in Figure P3-9.

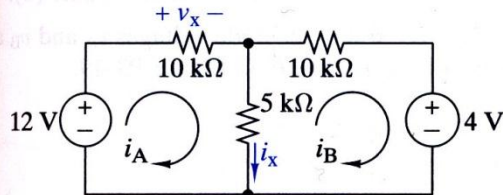


FIGURE P3-9

(b) Solve for i_A and i_B .

(c) Use these results to find v_x and i_x .

3-10(a) Formulate mesh-current equations for the circuit in Figure P3-10.

(b) Solve for i_A and i_B .

(c) Use these results to find v_x and i_x .

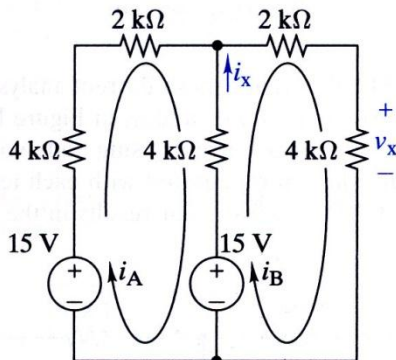


FIGURE P3-10

3-12(a) Formulate mesh-current equations for the circuit in Figure P3-12.

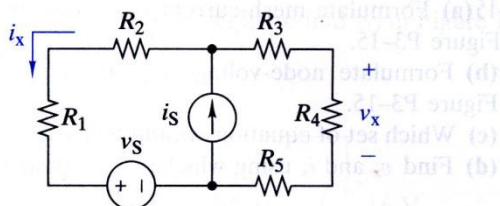


FIGURE P3-12

(b) Solve for v_x and i_x when $R_1 = 200 \Omega$, $R_2 = 300 \Omega$, $R_3 = 50 \Omega$, $R_4 = 250 \Omega$, $R_5 = 200 \Omega$, $i_s = 100 \text{ mA}$, and $v_s = 25 \text{ V}$.

(c) Find the total power dissipated in the circuit.

3-15(a) Formulate mesh-current equations for the circuit in Figure P3-15.

(b) Formulate node-voltage equations for the circuit in Figure P3-15.

(c) Which set of equations would be easier to solve? Why?

(d) Find v_x and i_x using whichever method you prefer.

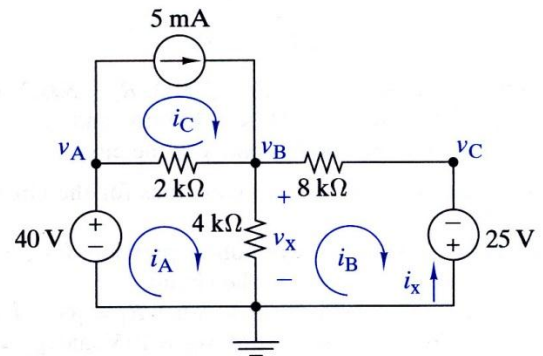


FIGURE P3-15

3-20(a) Use MATLAB and mesh-current analysis to solve for the mesh currents i_A , i_B , i_C , and i_D in Figure P3-20.

(b) Solve for the same currents using node-voltage analysis and compare the effort required with each technique.

(c) Use OrCAD to verify your results in the previous two parts.

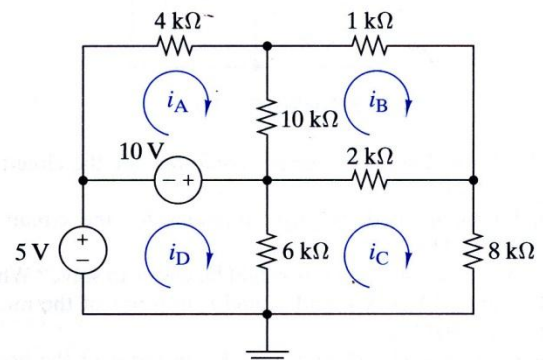


FIGURE P3-20

3-22 Find the proportionality constant $K = i_O/i_S$ for the circuit in Figure P3-22.

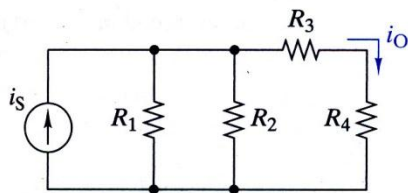


FIGURE P3-22

3-26 Use the superposition principle to find v_O in Figure P3-26.

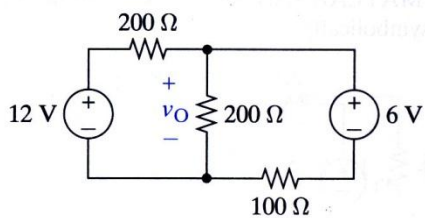


FIGURE P3-26

4-2 Find the voltage gain v_O/v_I and the current gain i_O/i_S in Figure P4-2. For $i_S = 2$ mA, find the power supplied by the input current source and the power delivered to the 2-kΩ load resistor.

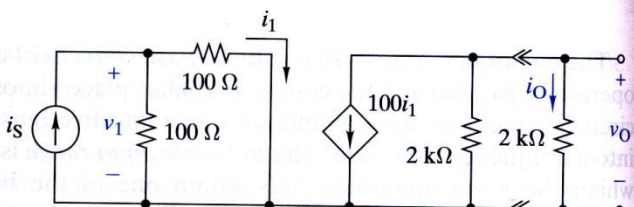


FIGURE P4-2

4-6 Find an expression for the current gain i_O/i_S in Figure P4-6. (Hint: Apply KCL at node A.)

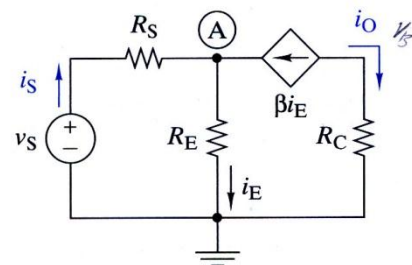


FIGURE P4-6

4-13 Find R_{IN} in Figure P4-13.

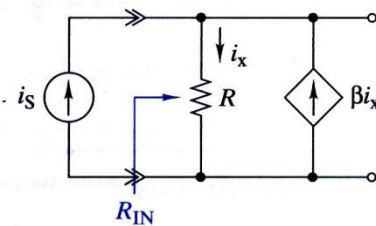


FIGURE P4-13